

BUILDING A PERSONALIZED BEHAVIORAL NUDGE FRAMEWORK FOR ATRIAL FIBRILLATION CARE

Arshia Thota | HelloAlfred.AI | Informatics, Capstone Fall 2026

Supervisor: Dr. Ajay Tripuraneni

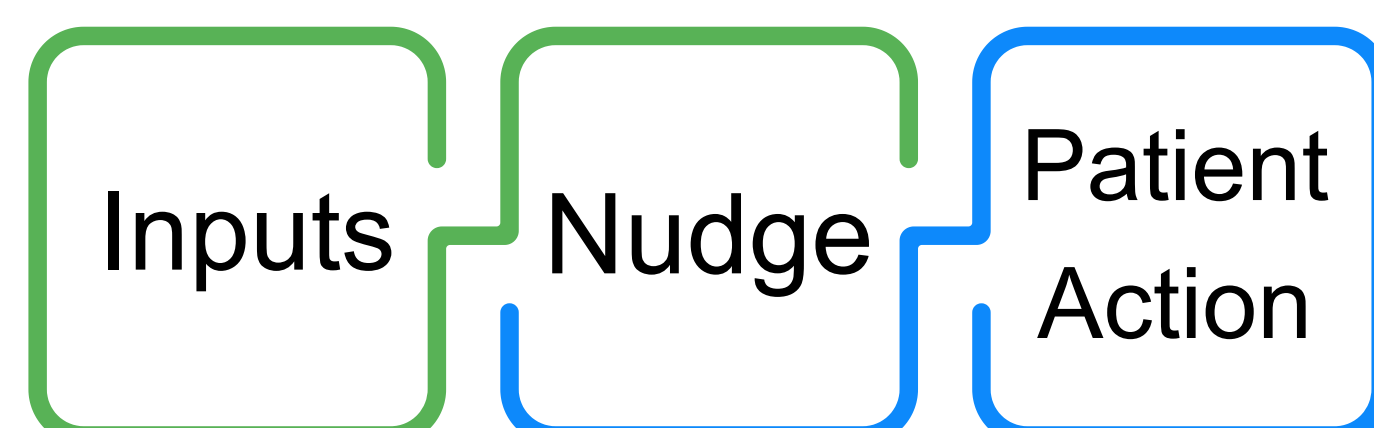
Motivation



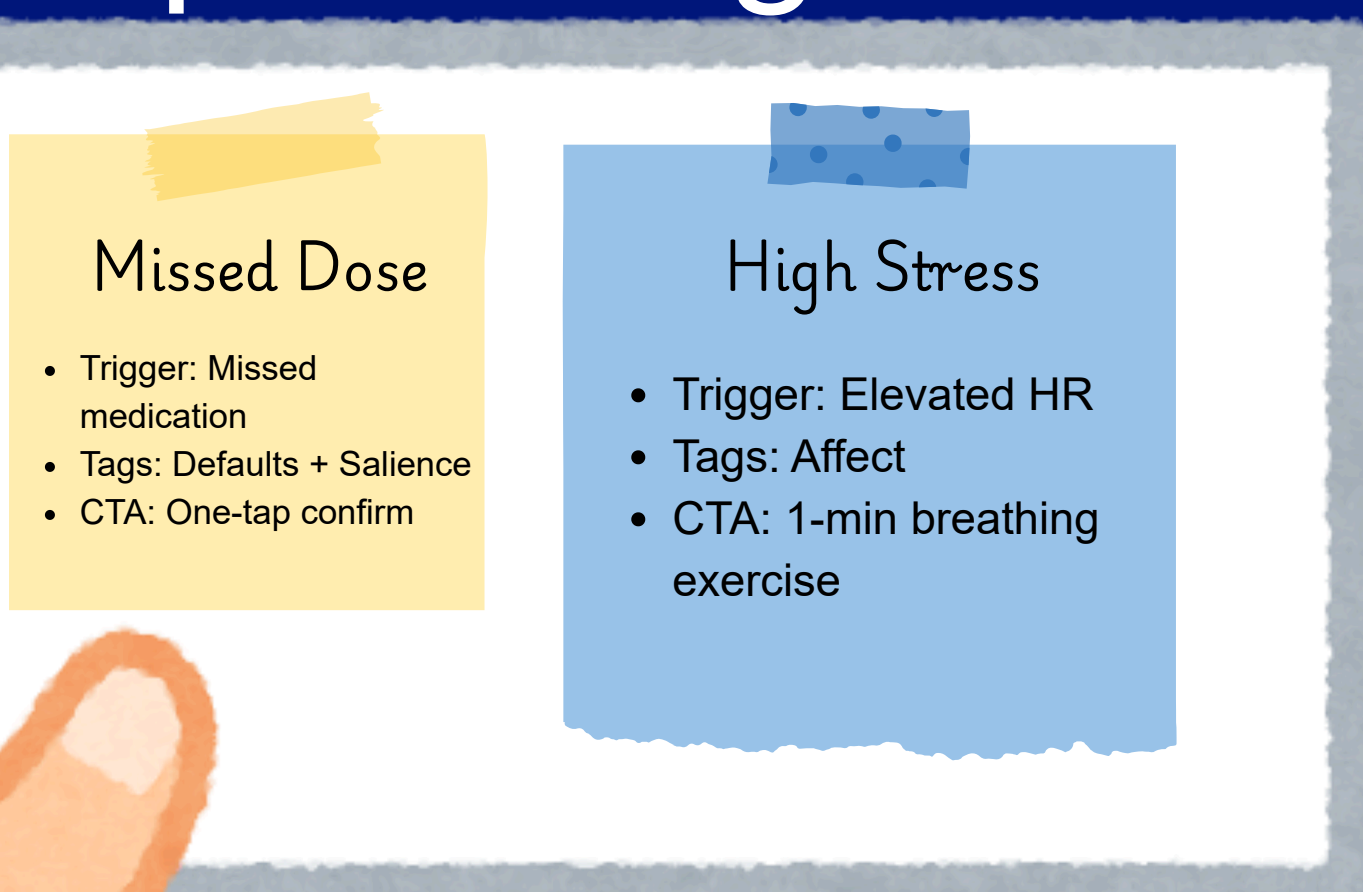
- AF patients struggle with medication adherence and follow-up tasks.
- Engagement varies by capability, opportunity, and motivation.
- HelloAlfred needs a behavior-aware system that adapts to each patient.

Goal

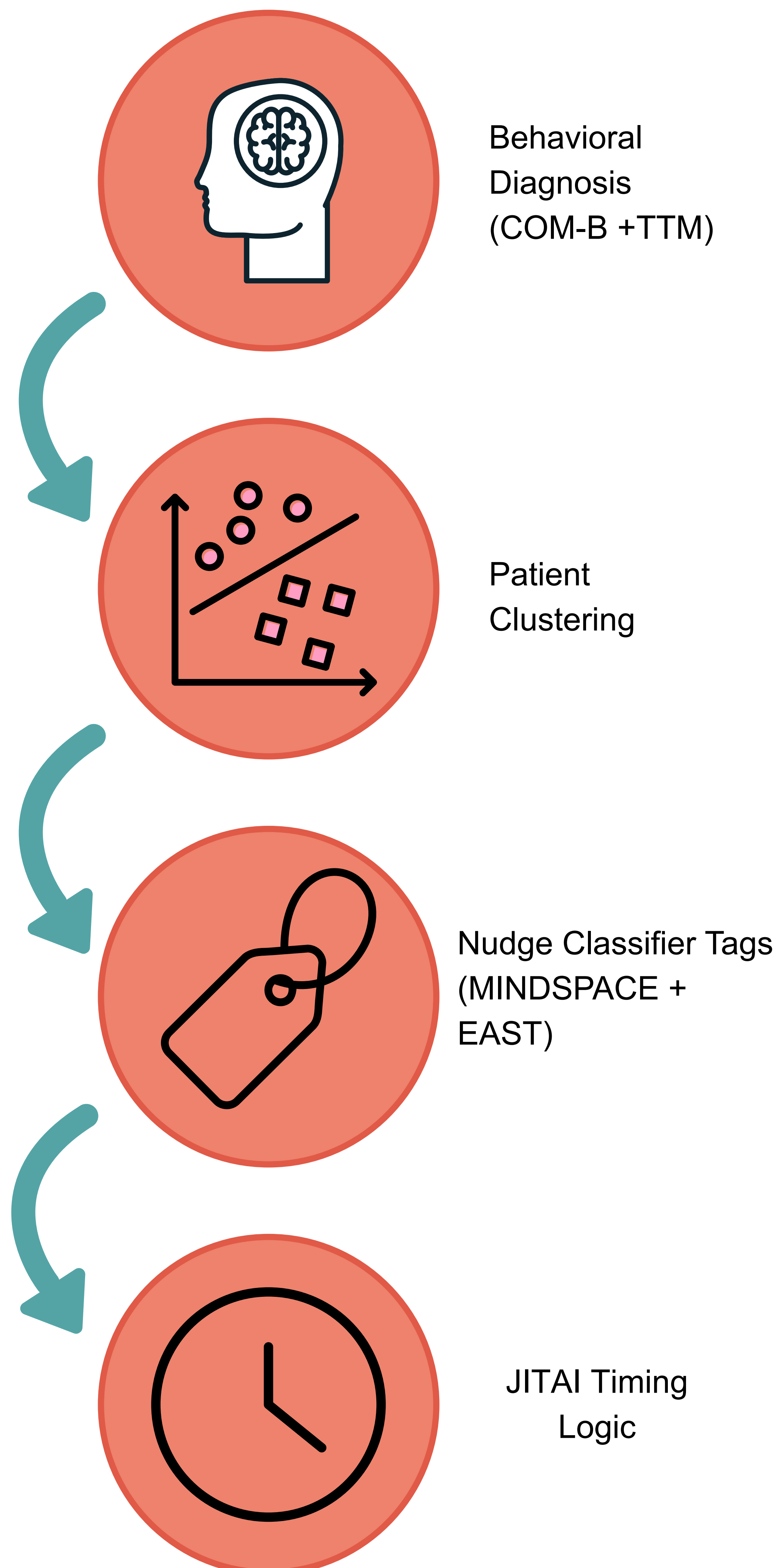
Develop a personalized nudge framework that identifies behavioral barriers and delivers adaptive, timely support to help AF patients stay on track with their care.



Example Nudges



Methodology



Final Product

Before

Reminder	
• One-size-fits-all nudges • No behavioral segmentation • Static reminders	
Accept	Decline

After

Reminder	
• Personalized nudges matched to COM-B needs • Cluster-sensitive content • Event-triggered JITAI timing • Ethical, transparent nudge specs	
Accept	Decline

Conclusion

- Created a unified nudge framework combining behavioral science + ML.
- Built synthetic patient simulations to test nudge logic safely.
- Framework will inform AF patient engagement product features at HelloAlfred.
- I really enjoyed working on this project and learning how behavioral models can shape real patient support. It pushed me to connect research, data, and design in a way that felt meaningful.